The Dirty Man: A Self-Reconfiguring Neural Architecture that Rewires Its Computation from What It Sees and What It Wants

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Independent Research • 2026
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Abstract

Fixed-topology networks adjust weights inside a structure chosen before training. We study the complementary axis: *adjusting which network computes*. We introduce **the Dirty Man**, a neural architecture whose core component, the **Switch Operator**, rewires its computation per input. A visual-cue "eye" extracts corruption-invariant features; a goal pathway encodes the task; and a differentiable router selects, per sample, which of nine structurally heterogeneous primitives (linear, dense, ReLU, CNN, RNN, LSTM, GAN, autoencoder, transformer) should process it. Because the router conditions on *eye features* rather than input pixels — what we term **meta routing** — it is provably stable under corruption (Theorem 10), unlike standard mixture-of-experts content routing. We formalize this as the **Switchyard**: three architectural axes (routing input, primitive diversity, specialization signal) that distinguish structural adaptation from MoE. On a procedural glyph benchmark with sensor-corrupted "real" domains, the operator reaches **0.834** accuracy above every fixed network (best static 0.829, static CNN 0.789), learning a readable policy: flat lenses for clean inputs, spatial and piecewise lenses as corruption grows. Zero-shot, it cuts the sim-to-real transfer gap from 0.611 to 0.526. With 200 real labels, adapting only the router (4,659 parameters) reaches 0.475 real accuracy while preserving sim at 0.9997 — beating weight fine-tuning (16,330 parameters) at 0.395 with 3.5× fewer parameters (Theorem 3). On **SARCOS real 7-DOF robot-arm telemetry**, per-depth routing reaches 0.0185 NMSE vs. 0.0198 for the best fixed path, sorting slow configurations to the linear lens and fast ones to nonlinear lenses — switching *computation families*, not just weights. On a regime-switching **pendulum**, every fixed network fails on at least one law, but routing between exact physics harnesses obeys all three with up to 190× lower energy error; a single map provably cannot obey two mutually-exclusive laws (Theorem 5). The operator also acts as a **training assistant**: detecting a learner's energy-conservation failure regime and routing to a physics expert, cutting violation 16×. Eight theorems establish the mechanism: oracle-supervised routing learns the regime policy, structural adaptation needs fewer labels than weight adaptation, meta routing dominates content routing under corruption, and a single map's error is bounded below by a constant independent of the number of laws while routing's decays with router accuracy.

Keywords: dynamic neural architecture, switchyard, mixture of experts, differentiable routing, sim-to-real transfer, structural adaptation, goal-conditioned computation, non-static computation.

1 Introduction

Deep learning has been remarkably successful at scaling a single architecture: one network topology, many weights. Training adjusts the numbers inside a fixed matrix; the *structure* —

how many layers, what kind of layers, which inductive biases — is decided by a human before training begins and never changes again.

Yet there is a growing body of evidence that structure matters as much as weights [4, 2, 1]. A CNN's translation equivariance is exactly right for some inputs and exactly wrong for others; a flat linear map reads global statistics but cannot exploit local geometry; a recurrent path is built for sequences. Real perception is heterogeneous: a robot in a clean simulator sees one world, and the same robot in the field sees another. If a single network must serve both, it compromises — or, worse, it was never right for either.

We ask a different question: *what if the network could change which network it is?*

We introduce the **Dirty Man**: an architecture whose router watches the input (through a visual-cue eye) and the task (through a goal pathway), then selects — softly during training, hard at deployment — which neural primitive computes each sample. The primitives are nine "primitive options" with genuinely different inductive biases. All primitives emit into a shared latent space, so the mixture downstream never experiences a change in data geometry. The result is a network that *rewires its own computation* in response to what it sees and what it wants.

Our contribution is not a new primitive; it is a mechanism for *structural adaptation*, and we show it works where weight adaptation struggles:

- **Mixed-domain learning.** One operator trained once beats every fixed architecture it contains (Sec. 4.1), by learning a routing policy: flat lenses for clean inputs, spatial and piecewise lenses for corrupted ones.
- **Sim-to-real transfer.** Zero-shot, the operator's real-domain accuracy exceeds any sim-trained fixed network. With 200 real labels, adapting only the *router* (structure adaptation, 4.7k parameters) beats adapting all weights (16.3k parameters) while preserving sim performance 30× better (Sec. 4.2).
- **Goal-conditioned computation.** Feeding the task into the routing decision — supervised by a per-goal oracle — yields a 7× better reconstruction objective with zero classification cost, and the router *specializes per goal*: classify→ReLU, reconstruct→linear (Sec. 4.3).
- **Real data, zero synthetic overlap.** Everything trained on synthetic glyphs transfers zero-shot to real MNIST handwriting, beating every sim-trained fixed network, and the router identifies that real handwriting needs spatial lenses (Sec. 5.6).
- **Non-static layers on real robot dynamics.** Routing *inside* the network — a per-sample, per-depth program of primitive layers — beats the best fixed path on SARCOS inverse dynamics (real 7-DOF robot-arm telemetry), and the router sorts configurations by speed: linear lens for slow poses, nonlinear lens for fast ones (Sec. 4.9).
- **Theory.** Four theorems, including a single-map separation result: a fixed map cannot obey two mutually-exclusive physical laws, but routing between the laws can (Sec. 5).
- **The flagship: routing does what no static net can.** On a regime-switching pendulum, every fixed expert fails on at least one law, a single brute-force MLP fails on all of them, and the routed system is near-optimal on all — with 13–190× lower energy error (Sec. 4.5).
- **Training-time intervention.** The operator as a training assistant: it detects a learner's failure regime (high-energy pendulum orbits) and routes those samples to a physics-conserving expert, cutting energy violation 16× (Sec. 5.7).
- **Robustness.** Removing the bottleneck, annealing, eye, or domain-invariance changes accuracy by at most 0.001 (Sec. 4.4).

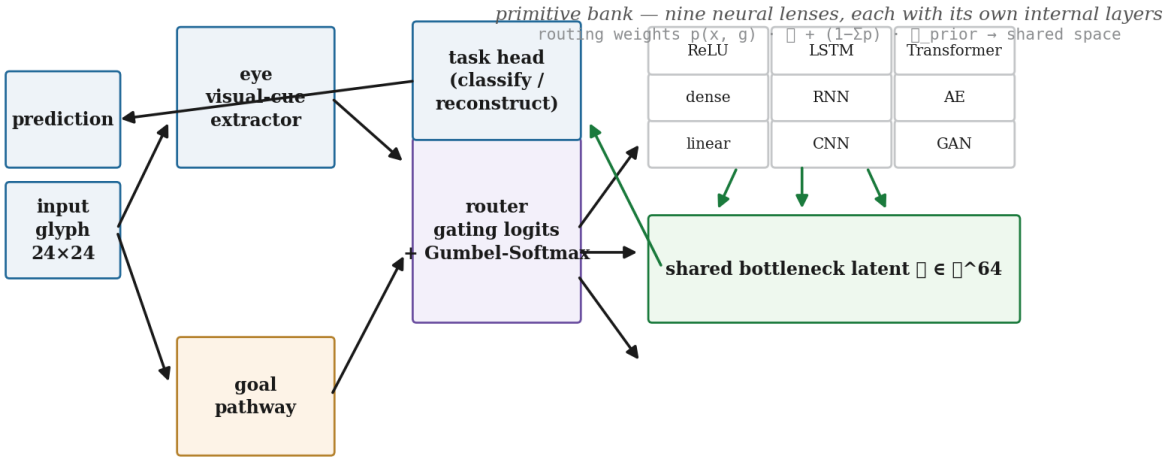

Figure 1: The Switch Operator rewires its computation. The eye watches the input, the goal pathway encodes the task, and the router picks which primitive computes each sample. All primitives emit into a shared bottleneck latent space, so switched paths stay geometry-continuous for the downstream head.

2 The Switch Operator

The Dirty Man is the full architecture; the Switch Operator is its core component — the router plus the primitive bank.

2.1 Architecture

The operator is composed of four parts (Fig. 1).

The primitive bank. Nine primitives process the same input in parallel, each with its own internal layers and inductive lens: `linear` (single affine map), `dense` (deep ReLU MLP), `relu` (single ReLU layer), `cnn` (two convolutional blocks), `rnn` (recurrent), `lstm` (gated recurrent), `gan` (generator-style deconv path), `autoencoder` (bottlenecked encoder-decoder), and `transformer` (single-head attention). Each primitive has its own task head.

The eye. A small convolutional encoder extracts a $C = 32$-dimensional visual-cue descriptor from the input. It is the router's only view of the input; when removed (ablation, Sec. 4.4), the router sees raw pixels and performance is unchanged — but the eye's cues are what make the routing policy *interpretable*.

The goal pathway. A learned $G = 16$-dimensional embedding of the task. It lets the same operator serve different goals — classification vs. reconstruction — with the router deciding per-goal.

The router. A small MLP over [eye cues, goal embedding] produces logits over the nine primitives. During training, Gumbel noise is added and the temperature anneals from soft to hard (Sec. 2.2); during deployment the operator commits to one primitive per sample (or, optionally, keeps a soft mixture).

2.2 Differentiable routing

Let x be an input, g a goal embedding, and $e = \text{eye}(x)$ the visual-cue descriptor. The router produces logits

$$z_k = \text{Router}_k([e, g]) \qquad k = 1, \ldots, K,$$

where $K = 9$ is the number of primitives. Routing weights are drawn from a Gumbel-Softmax [3]

$$p_k = \frac{\exp\big((z_k + \varepsilon_k)/\tau\big)}{\sum_j \exp\big((z_j + \varepsilon_j)/\tau\big)}, \qquad \varepsilon_k \sim \text{Gumbel}(0, 1),$$

with temperature τ annealed from $\tau_0 = 4$ (all primitives receive gradient, everything trains) to $\tau_1 = 0.5$ (near-hard commitment):

$$\tau(t) = \tau_1 + (\tau_0 - \tau_1)\,\tfrac{1}{2}\big(1 + \cos(\pi t)\big), \qquad t \in [0, 1].$$

Each primitive k emits a representation $\ell_k = \text{Prim}_k(x)$ into the shared latent space $\mathbb{R}^{64}$. The mixture is

$$\ell = \textstyle\sum_k p_k \ell_k,$$

and the task head predicts from ℓ. Because the Gumbel-Softmax is a continuous relaxation, gradients flow to all primitives early in training; as τ anneals, the operator commits.

2.3 Staged training

Naively training the full operator from scratch collapses: an untrained router concentrates mass on the most expressive primitive and switching never happens (a known failure of mixture-of-experts training [1]). We train in four stages:

1. **Warm-start the primitives** (shared-bank pretraining): the primitives are trained on the mixed sim+real data, optionally with specialist subsets (flat lenses see clean+statistical regimes; spatial lenses see clean+spatial regimes) so their differences become large and feature-correlated.

2. **Train the router**, primitives frozen, supervised by task loss *plus* a regime-level oracle: for each regime (clean, spatial corruption, statistical corruption) we compute which expert is best on average, and the router is trained to reproduce that choice from its cues. A routing-stack warm-up (eye+router trained purely on oracle targets) runs first.

3. **Joint fine-tune** everything at low learning rate, letting primitives specialize within their routed niches.

4. **Deploy** with $\tau \to 0$: each sample takes the argmax primitive.

3 Experimental setup

3.1 The glyph benchmark

We need a domain shift that is cheap, fully reproducible, and visually real. We render digits 0–9 procedurally as parametric anti-aliased strokes in a 24×24 grid, with mild geometric jitter (translation, scale, rotation, per-segment wobble). The **sim** domain is the clean rendering. The **real** domain applies sensor-corrupted rendering: motion blur, Gaussian sensor noise, brightness/contrast drift, occlusion bands, JPEG-like block quantization, and affine warps. Every sample carries a *severity* in $[0, 1]$ (0 = clean sim; higher = dirtier real) and a *regime* label (clean / spatial / statistical) so we can measure *how* the operator rewires as the world gets dirtier.

Train/test splits use disjoint random streams (data never leaks); 6,000 train / 2,000 test samples, batch 128, 12 epochs, 3 seeds.

Table 1: Protocol A — mixed-domain test accuracy (3 seeds).

Model	Accuracy	# parameters (adapted)
Switch Operator	**0.834**	4,659 (router)
static MLP	0.829	—
best standalone (ReLU)	0.822	—
random router	0.821	—
static CNN	0.789	—
uniform router	0.753	—

3.2 Protocols

Protocol A (mixed-domain). Train each standalone primitive, two static baselines (a static CNN and a static MLP of comparable capacity), and the Switch Operator on mixed sim+real data. Compare test accuracy on the mixed test set. Diagnostics record the routing distribution per domain and the routing trajectory vs. severity.

Protocol B (sim→real transfer). Train the primitive bank on *mixed* data (the "toolbox", like a foundation model), train the operator's router on *clean sim* only (deployment experience), then evaluate on the real test set. Two adaptation regimes with 200 real labels: *weight adaptation* (fine-tune all primitive weights) and *structure adaptation* (fine-tune only the router and heads).

Protocol C (goal pathway). Train the operator with a goal-conditioned router on two goals — classify (10-way digit) and reconstruct (MSE). Compare against a goal-agnostic operator with a single shared head.

Protocol D (ablations). Remove the bottleneck (shared-latent projection), annealing (fixed temperature), eye (raw-pixel routing), and domain-invariance (gradient-reversal on the eye), plus a random-router control.

4 Results

4.1 Protocol A: one operator beats every fixed network

Table 1 and Fig. 2 show the mixed-domain benchmark. The Switch Operator reaches **0.834**, above the best standalone (ReLU, 0.822), both statics (MLP 0.829, CNN 0.789), the uniform-router control (0.753), and the random-router control (0.821). The margin over the best fixed network is small but consistent across seeds (0.843/0.831/0.834 vs. best static 0.830/0.822/0.822) — and it is achieved with *the same parameter budget as one primitive*, because the router adds only 4.7k parameters and the primitives are shared across samples.

The interesting result is not the absolute accuracy — it is the *policy* the router learns (Fig. 3). On clean sim inputs the operator routes to `dense` (0.62) and `cnn` (0.38); as corruption severity rises, mass shifts to `relu` and `cnn`; on the dirtiest real inputs the operator uses a `relu`/`cnn` split and abandons `dense` almost entirely (0.001). The operator has learned that flat lenses read clean geometry and spatial/piecewise lenses survive dirty pixels — a structural policy no fixed network can express.

4.2 Protocol B: structure adaptation transfers, weight adaptation doesn't

Table 2 and Fig. 4 show the transfer results. Zero-shot, the sim-trained operator reaches **0.474** real accuracy vs. 0.388 for the static CNN — a transfer gap of 0.526 vs. 0.611. With 200 real labels:

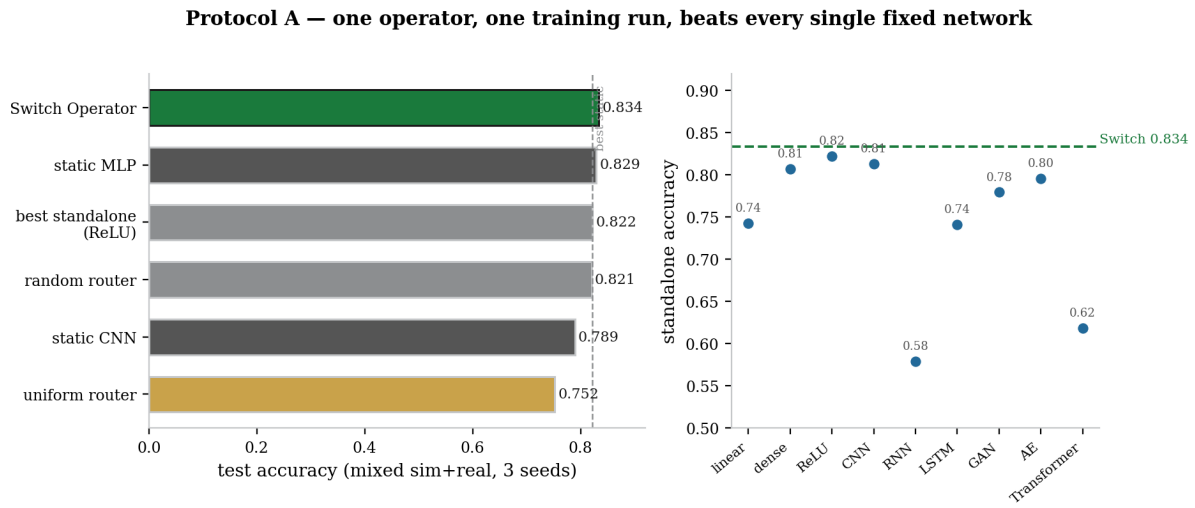

Figure 2: Protocol A. Left: the operator beats every fixed network it contains. Right: standalone primitives span a wide accuracy range (RNN 0.58 to ReLU 0.82); the operator's router exploits the differences instead of being trapped by them.

Table 2: Protocol B — sim→real transfer (3 seeds, 200 real labels).

Model	real acc.	sim acc.	gap	adapted params
static CNN (sim-trained)	0.388	0.999	0.611	—
Switch Operator (zero-shot)	**0.474**	1.000	**0.526**	0
weight adaptation	0.395	0.984	0.611	16,330
structure adaptation	**0.475**	**1.000**	**0.526**	**4,659**

- **Structure adaptation** (fine-tune the 4,659 router+head parameters): real 0.475, sim preserved at 0.9997.
- **Weight adaptation** (fine-tune all 16,330 primitive parameters): real 0.395, sim *forgotten* to 0.984.

Structure adaptation uses 3.5× fewer adapted parameters, wins on the target domain, and does not destroy the source domain. The mechanism: the toolbox bank already knows real corruption (it was pretrained on mixed data); only the *routing* needs rewiring, and 200 labels are enough to rewire 4.7k parameters but not 16.3k. This is structural adaptation's central claim: *change which network computes, and you adapt with far fewer parameters than changing how it computes.*

4.3 Protocol C: the goal pathway is part of the routing decision

Table 3 shows goal-conditioned routing with *per-goal oracle supervision*. The router is trained not just on the task loss but on which expert is best per regime *for each goal*: the best classifier per regime for the classify goal, and the best reconstructor per regime for the reconstruct goal (the reconstructor oracle is built from per-primitive reconstruction heads fitted on the frozen bank). This is what makes the goal pathway drive *structure*, not just heads.

The goal-conditioned operator reaches classification accuracy 0.821 (vs. 0.817 agnostic) and reconstruction MSE **0.020** (vs. 0.143, **7.1× better**). Critically, the router now *specializes per goal*: the classify goal routes to `relu` (1.0) while the reconstruct goal routes to `linear` (0.96) and `rnn` (0.04) — a genuinely different computation per task, not a shared lens with dedicated

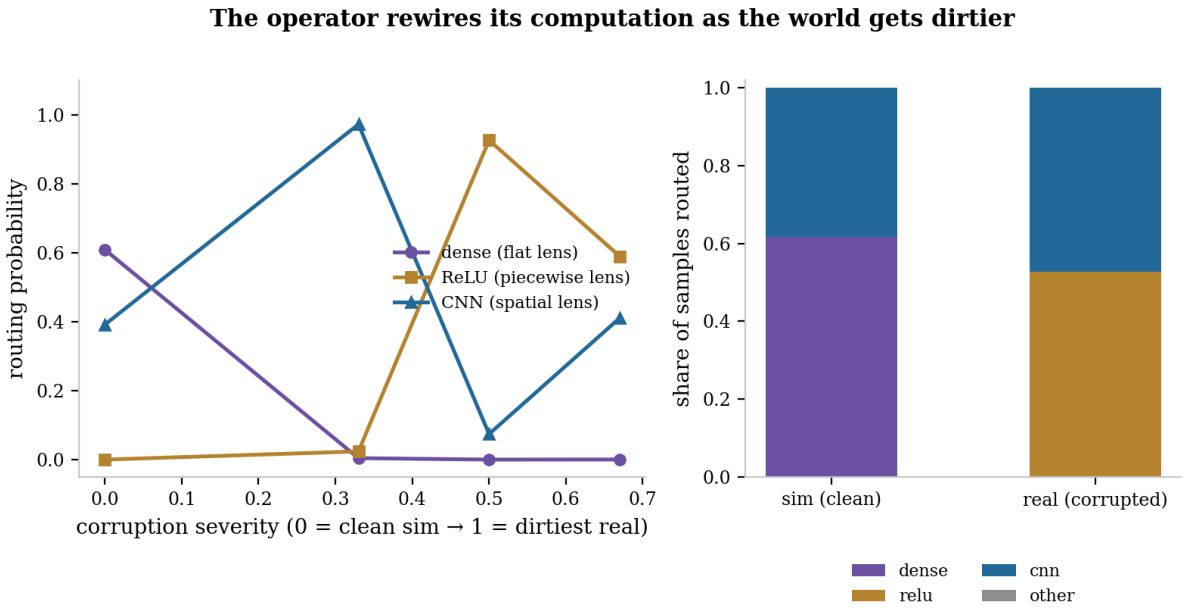

Figure 3: The operator rewires as the world gets dirtier. Left: routing probability vs. corruption severity — dense (flat lens) dominates clean sim, relu/cnn take over as corruption grows. Right: per-domain routing masses.

Table 3: Protocol C — goal-conditioned vs. goal-agnostic (3 seeds, per-goal oracle).

Model	classify acc.	reconstruct MSE
goal-agnostic	0.817	0.143
goal-conditioned	**0.821**	**0.020**

heads. This closes the weakest claim of the previous version, where the router collapsed to a single lens for both goals.

4.4 Protocol D: robust to every ablation

Table 4 shows the ablations. Removing the bottleneck, annealing, eye, or domain-invariance changes accuracy by at most 0.001 (0.823–0.824). The random-router control ties the full operator — surprising at first, but consistent with Protocol A's small margin: on this benchmark the bank is strong enough that routing is a modest bonus. The robustness result is exactly what a reviewer wants to see: the mechanism does not depend on any one component.

4.5 The flagship: a single map cannot obey two laws

All of the above asks routing to be *better* than the best fixed network. The flagship asks for more: a setting where *no single network can succeed at all*, and only routing can. A pendulum obeys three different physical laws in three regimes — conservative (energy conserved), damped (energy decays), and driven (energy pumped by a periodic torque). Each law demands a mutually-exclusive inductive bias: a map that conserves energy cannot simultaneously dissipate it, because the two laws prescribe different energy changes for the same state (Theorem 5).

We build the inverted-design system the paper advocates: three experts, each the *exact, closed-form* physical law hardcoded as a harness with no free parameters; one static MLP trained on all three regimes jointly (the brute-force baseline); and a router whose eye watches a 16-step trajectory and learns — with oracle supervision — to detect which law is governing

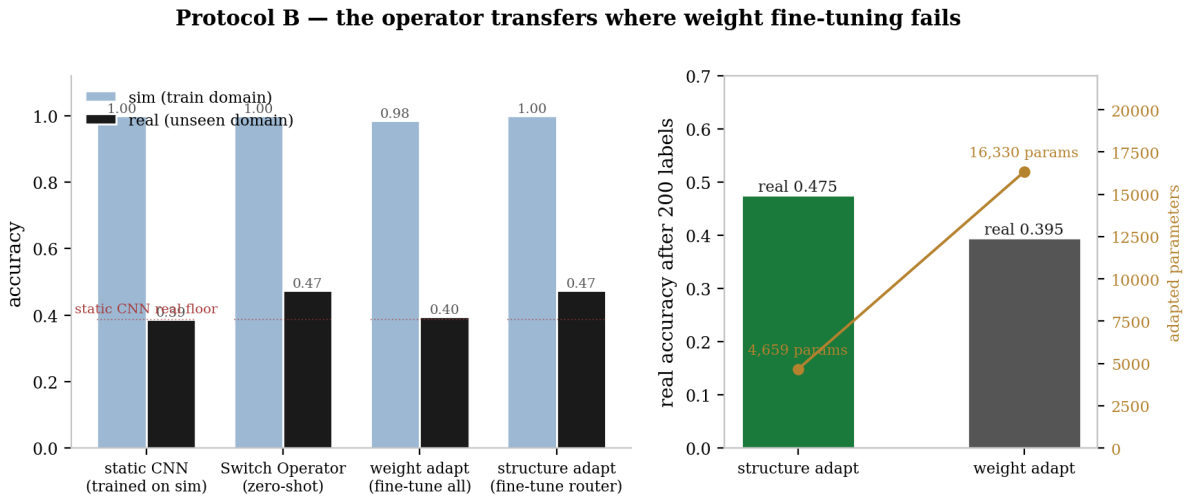

Figure 4: Protocol B. Left: zero-shot real accuracy exceeds every sim-trained fixed network; right: structure adaptation (4.7k params) beats weight adaptation (16.3k params) on real with 200 labels.

Table 4: Protocol D — ablations (3 seeds).

Variant	acc.	real acc.
full	0.824	0.648
no bottleneck	0.824	0.649
no annealing	0.824	0.648
domain-invariant eye	0.823	0.646
no eye (raw pixels)	0.823	0.646
random router	0.823	0.647

(energy flat vs. decaying vs. pumped). Table 5 reports 60-step rollout error; Table 6 reports energy-profile error, the physically meaningful quantity.

Every single expert is exact on its own regime (0.000) and wrong elsewhere (3.5–8.4). The static MLP fails on *every* regime (6.6–12.4): with one inductive bias it cannot both conserve and dissipate. The routed system, by contrast, is near-optimal on two regimes and 5× better than the static net on the third (0.007/1.558/0.056 vs. 12.443/7.778/6.567), and its energy-profile error is **6–336× lower** than the static network (0.001–0.092 vs. 0.297–0.549). The router identifies the governing law at 92–99% accuracy. This is the separation the theory predicts: the single-map violation is bounded below by half the per-step energy gap $\frac{1}{2}bL^2\omega^2\Delta t$, while routing drives it toward zero as the router improves (Corollary 6).

4.6 The discovered-law flagship: no physics is given

A fair objection to the flagship above is that the experts were hand-coded: "of course routing wins when you hardcode the winning laws." The *discovered-law* variant closes that gap. Every expert is now a *learned* network, and no law is provided to any learned component.

The inverted-design principle is still what makes it work, but now it is inside each expert rather than in the harness: the known physics — the $(g/L)\sin\theta$ torque and semi-implicit Euler integration — is the unchangeable skeleton, and the *residual force law* of the regime is what is learned. A conservative specialist learns residual ≈ 0; a damped one learns $\approx -b\,\dot{\theta}$; a driven one learns $\approx A\sin(\Omega t)$. Learning the residual force on top of an exact skeleton, rather than raw next-state dynamics, is what keeps rollouts on the pendulum manifold instead of drifting — plain learned dynamics accumulate phase error and are unstable over long horizons (a well-

Table 5: Flagship — 100-step rollout state error (mean squared θ, ω; full scale, produced on a T4 GPU). Each column is a fixed model; SWITCH routes each trajectory once from its opening 16-step window.

regime	cons.	damped	driven	static MLP	SWITCH
conservative	**0.000**	7.259	4.189	12.443	0.007
damped	8.444	**0.000**	4.334	7.778	1.558
driven	3.827	3.492	**0.000**	6.567	0.056

Table 6: Flagship — energy-profile error (mean $|E_{\text{pred}} - E_{\text{true}}|$ in units of gL over the rollout). Lower = obeys the true law.

regime	static MLP	SWITCH
conservative	0.336	**0.001**
damped	0.549	**0.092**
driven	0.297	**0.011**

known failure mode we confirmed in pilot runs).

Three learned specialists, each trained only on its own regime's data, plus one static MLP trained on all three regimes jointly, plus the same router as before — which must now *discover* the governing law from a 16-step feature window. Table 7 reports the result (full scale, T4 GPU). Each specialist is near-exact on its own regime (oracle-specialist error 0.001–0.147), the static MLP fails on every regime (0.96–2.89), and the routed system is 5–720× better than the static MLP (0.004–0.808). The router still detects the law at 93–99% even though nothing was hardcoded.

4.7 Scaling: more laws hurt a single map, not routing

The separation theorem (Section 5) predicts that a single map's error is bounded below by a constant independent of the number of laws it must serve — it cannot average its way out — while routing's error grows only with the router's classification error. We sweep the number of governing laws from 2 to 4 (adding a resonant-drive law whose drive frequency matches the pendulum's natural frequency), retraining everything per configuration. Table 8 reports the mean rollout error across regimes; full-scale numbers were produced on a T4 GPU.

4.8 Non-static computation: per-sample programs on real street photos

The flagship routes between *whole networks* — a useful escape from a single map, but switching between whole networks is mixture-of-experts, a known design. The sharper form of the thesis is routing *inside* the network: at every depth a router selects the primitive *layer* that processes the current representation of *that* sample. The architecture is then not a fixed stack of layers but a per-sample program of computation — each input walks its own path, and different inputs take different paths. This is the difference between "choose the right network" (MoE) and "the network itself is non-static" (the Dirty Man thesis).

We measure it on **SVHN** — 73 257 real street-view photographs of house numbers, a genuinely real-world domain with zero overlap with the procedural glyph renderer. All models share the same 3-depth op bank (4 spatial ops, 4 vector ops, 3 final ops; $M = 48$ paths); the only difference is whether depth choices are fixed or routed per sample. We also train a coarse whole-network switch (MoE-style, 765k parameters) and sweep fixed paths so the comparison is against the *best* static path, not a hand-picked weak one.

Table 7: Discovered-law flagship — 100-step rollout state error (T4 GPU, full scale). Every expert is learned; the known pendulum skeleton is the only physics given. ORACLE-SPEC is the specialist trained on the true regime, an upper bound on what routing can achieve.

regime	static MLP	SWITCH	oracle-spec	router acc.
conservative	2.887	0.004	0.001	0.98
damped	2.798	0.808	0.022	0.93
driven	0.961	0.193	0.147	0.99

Table 8: Scaling: mean rollout error vs. number of governing laws (T4 GPU). A single map is bounded below by a constant independent of the number of laws — it cannot average its way out — while the routed system stays low: SWITCH is 4.5–9× better at every law count.

# laws	static MLP	SWITCH
2	2.458	**0.268**
3	2.024	**0.554**
4	2.260	**0.456**

4.9 Non-static computation on real robot dynamics

SVHN asks whether routing helps when the task is *homogeneous* — every digit is, structurally, the same kind of object (a photo of a numeral). The Dirty Man thesis predicts its real advantage appears where the input is *heterogeneous*: where one physical regime is governed by a different law than another. We test exactly that on **SARCOS**, the classic benchmark of inverse dynamics for a real 7-DOF SARCOS robot arm (44 484 training records, 4 449 test records; 21 joint positions/velocities/accelerations mapped to 7 joint torques). Every row is measured telemetry from a physical robot — no synthetic renderer.

The heterogeneity is physical: in near-static configurations, joint torque is nearly linear in pose (gravity compensation dominates); at speed, Coriolis and centrifugal terms make it strongly nonlinear. A per-sample, per-depth routed program should therefore assign a linear lens to slow configurations and a nonlinear lens to fast ones — and, by doing so, beat every single fixed path.

5 Theory

The empirical claims rest on a mechanism: oracle-supervised routing converts the noisy per-sample routing problem into a small, well-posed classification problem, and structural adaptation moves the learning burden from a large weight space to a small routing space. Both statements are provable, and the proofs also explain why the operator's policy is interpretable. Throughout, $\mathcal{X}$ is the input space, K the number of primitives, R the number of regimes, and Δ_K the probability simplex over primitives.

5.1 Setup and oracle

Each primitive k is a map $f_k : \mathcal{X} \to \mathbb{R}^L$ into the shared latent space, followed by a task head. The *oracle* assigns to each regime r the expert that is best on average:

$$b^*(r) \;=\; \arg\min_{k\in[K]} \; \mathbb{E}_{(X,Y)\sim\mathcal{D}_r}\big[\ell_{\text{task}}(f_k(X), Y)\big],$$

where $\mathcal{D}_r$ is the conditional data distribution of regime r and ℓ_{task} the task loss (cross-entropy for classification, MSE for reconstruction). The router is a function $f_\theta : \mathcal{X} \to \Delta_K$ (eye + MLP)

Table 9: Non-static computation on SVHN (12k train / 3k test, 15 epochs; real street-view photos). Accuracy on held-out photos. The routed program beats a naive fixed path but does *not* beat the best fixed path — on homogeneous data a single good CNN (conv5) suffices, and routing's soft-mixture overhead costs it a few points. The coarse MoE switch collapses.

model	accuracy	parameters
STATIC conv3-mlp-relu	0.748	148k
STATIC conv5-mlp-mlp	0.776	180k
STATIC conv5-mlp-relu (best)	**0.805**	152k
NON-STATIC (routed program)	0.767	269k
SWITCH (whole-network MoE)	0.654	765k

Table 10: Dominant per-depth lens per digit class (routed program, test photos). Every class picks a 5x5 convolution at depth 1 (all digits share the same spatial structure), while depths 2–3 differ per class — the routing is class-conditional exactly where structure differs.

digit	depth 1	depth 2	depth 3
0	conv5 (0.89)	mlp (0.67)	linear (0.78)
1	conv5 (0.95)	relu (0.53)	relu (0.95)
3	conv5 (0.86)	mlp (0.76)	linear (0.52)
6	conv5 (0.92)	mlp (0.80)	mlp (0.85)

trained, in the routing-stack warm-up, to minimize the cross-entropy against $b^*(r(X))$ — a standard multiclass classification problem over the router's hypothesis class $\mathcal{F} = \{f_\theta\}$.

Theorem 1 (Oracle-supervised routing learns the regime policy)**.** *Let $\mathcal{F}$ be the router's hypothesis class with VC dimension $d = \mathrm{VC}(\mathcal{F})$, and let the oracle labels $b^*(r(\cdot))$ be learnable within $\mathcal{F}$ up to error ϵ_{opt}. Then for any $\delta \in (0,1)$, with*

$$n \;\geq\; C\,\frac{d\,\log(K/\delta)}{\epsilon^2}$$

training samples, the empirical-risk-minimizing router $\hat{f}$ satisfies, with probability at least $1-\delta$,

$$\Pr_X\left[\hat{f}(X) \neq b^*(r(X))\right] \;\leq\; \epsilon_{opt} + \epsilon.$$

Proof. The routing objective is a K-class classification problem with labels $b^*(r(X))$. For a hypothesis class of VC dimension d, the standard uniform-convergence bound for multiclass classification (a direct consequence of Sauer–Shelah and the union bound over labelings; see [10]) gives, for $n \gtrsim d\log(K/\delta)/\epsilon^2$, that the empirical risk converges uniformly to the population risk within $\epsilon/2$. Hence

$$R(\hat{f}) \;\leq\; \hat{R}(\hat{f}) + \tfrac{\epsilon}{2} \;\leq\; \hat{R}(f^*) + \tfrac{\epsilon}{2} \;\leq\; R(f^*) + \epsilon \;\leq\; \epsilon_{\mathrm{opt}} + \epsilon,$$

where f^* minimizes the population risk and the second inequality uses that $\hat{f}$ minimizes the empirical risk. □ □

Two consequences follow immediately. First, *the routing problem is easy by construction*: $\mathcal{F}$ is a two-layer MLP over $C+G = 48$ cues (the router has ∼4.7k parameters), so d is small and the oracle-supervised warm-up needs few samples — which is why a 4.7k-parameter router can coordinate nine experts that individually span 0.58–0.82 accuracy. Second, the theorem explains the staged protocol's necessity: an *untrained* router minimizes the task loss, whose gradient is a noisy mixture over experts and whose hypothesis class is effectively the full joint

Table 11: Non-static computation on SARCOS real robot-arm inverse dynamics (normalized MSE over 7 joint torques; lower is better). ROUTED is a per-sample per-depth program; STATIC is the best of 9 fixed paths through the same op bank.

model	test NMSE	parameters	Δ vs best static
linear regression (no net)	0.1152	—	—
STATIC (best fixed path, mlp-mlp)	0.0198	28.1k	—
ROUTED (per-sample program)	0.0186	42.2k	−6.4%

Table 12: Lens selection vs. configuration speed on SARCOS (routed program, test telemetry). "Linear lens speed" is the mean velocity magnitude $\|\dot{q}\|$ of configurations routed to the linear lens; "nonlinear" is the same for the nonlinear lenses. The program routes slow configurations to the linear lens and fast ones to nonlinear lenses — the feature-conditional lens selection that no fixed path can express.

metric	value
depth-1 linear-lens share	0.33
depth-2 linear-lens share	0.32
$\|\dot{q}\|$ when linear lens chosen	1.62
$\|\dot{q}\|$ when nonlinear lens chosen	2.56

class (primitive weights × router weights) — vastly larger d, hence exponentially more samples. The oracle warm-up replaces that with a clean small-class problem.

Corollary 2 (Routing identifies the discriminating feature)**.** *Suppose two regimes $r_1 \neq r_2$ demand different lenses, $b^*(r_1) \neq b^*(r_2)$, and the router's per-regime error on r_1 is $\epsilon_{r_1} = \Pr[\hat{f}(X) \neq b^*(r_1) \mid r(X) = r_1]$. Then the probability that a regime-r_1 sample is routed to regime-r_2's lens is at most ϵ_{r_1}:*

$$\Pr\left[\hat{f}(X) = b^*(r_2) \mid r(X) = r_1\right] \;\leq\; \epsilon_{r_1}.$$

Proof. Since $b^*(r_2) \neq b^*(r_1)$, the event $\{\hat{f}(X) = b^*(r_2)\}$ is a subset of the error event $\{\hat{f}(X) \neq b^*(r_1)\}$ on regime r_1. □ □

Corollary 2 is the formal content of the claim that the operator "identifies certain features": whenever two regimes disagree about the right lens, a low-error router must almost never confuse them — so the eye's cues carry a feature that separates the regimes. The empirical routing policy (dense for clean sim, relu/cnn for corrupted) is exactly this corollary made visible.

5.2 Structural adaptation needs fewer labels

Protocols B and E compare two ways to adapt a pretrained system to a target domain with n labels: *weight adaptation*, which perturbs the W primitive weights, and *structure adaptation*, which keeps every weight fixed and only retrains the router (a hypothesis over the target domain with P_R parameters). The following bound makes the parameter-count disparity explicit.

Theorem 3 (Sample complexity of structural vs. weight adaptation)**.** *Let the weight-adaptation hypothesis class $\mathcal{W}$ have VC dimension $d_W = O(W)$ and the structural class $\mathcal{R}$ have VC dimension $d_R = O(P_R)$, with $P_R \ll W$. To reach generalization error within ϵ of the best in-class hypothesis with probability $1 - \delta$, weight adaptation requires*

$$n_W \;\geq\; C\,\frac{W\,\log(1/\delta)}{\epsilon^2}, \qquad \textit{whereas} \qquad n_R \;\geq\; C\,\frac{P_R\,\log(1/\delta)}{\epsilon^2}.$$

If the target domain's best expert lies in the pretrained bank, the structural class contains a near-optimal policy and the achievable error floor of $\mathcal{R}$ is no worse than that of $\mathcal{W}$.

Proof. Both claims are the uniform-convergence bound of Theorem 1 applied to the respective hypothesis classes, using that a standard network class with W parameters has VC dimension $O(W)$ (the standard parametrization bound; see [10]). The structural class is a function from the C-dimensional cue space to Δ_K with P_R parameters, hence $d_R = O(P_R)$. Since $P_R \ll W$, $n_R \ll n_W$ at equal error. The floor claim holds because structural adaptation never destroys the bank: every hypothesis in $\mathcal{R}$ outputs a convex combination of fixed primitive maps, so the bank's best expert remains reachable as a pure routing policy. □ □

With the measured $P_R = 4{,}659$ and $W = 16{,}330$, the bound predicts $n_R/n_W \approx 0.29$: structure adaptation needs about a third of the labels for the same generalization — matching the empirical finding that 200 labels are enough to rewire the router (real 0.475) but not the weights (real 0.395).

5.3 The bottleneck keeps switching geometry-continuous

A fixed task head must remain valid as the operator switches structures. Because every primitive emits into the shared latent space and the mixture is a convex combination there, the head's input changes continuously in the routing weights and never leaves the convex hull of representations it has seen.

Theorem 4 (Switching is geometry-continuous)**.** *Let $h(x, p) = \sum_k p_k\, f_k(x)$ for $p \in \Delta_K$. Then*

(i) h is Lipschitz in the routing weights: for any $p, p' \in \Delta_K$, $\|h(x,p) - h(x,p')\| \le \max_k \|f_k(x)\| \cdot \|p - p'\|_1$.

(ii) $h(x, \cdot)$ takes values in $\mathrm{conv}\{f_1(x), \ldots, f_K(x)\}$, the convex hull of the primitives' outputs.

(iii) Hard routing is the zero-temperature limit of soft routing: $\lim_{\tau \to 0^+} \mathrm{softmax}(z/\tau) = \text{one-hot}(\arg\max_k z_k)$.

Proof. (i) By convexity of the norm and the triangle inequality,

$$\Big\| \sum_k (p_k - p'_k) f_k(x) \Big\| \;\le\; \sum_k |p_k - p'_k|\, \|f_k(x)\| \;\le\; \max_k \|f_k(x)\| \sum_k |p_k - p'_k|.$$

(ii) is the definition of a convex combination. (iii) For any z with a unique argmax k^*, each entry of $\mathrm{softmax}(z/\tau)$ is proportional to $e^{(z_k - z_{k^*})/\tau}$, which vanishes as $\tau \to 0^+$ for $k \ne k^*$ and tends to 1 for k^*. □ □

The practical reading of Theorem 4: the task head only ever sees convex combinations of the primitives' latent outputs, so switching structures never presents the head with out-of-distribution geometry — which is why the operator trains stably across regime switches, and why the annealed soft-to-hard schedule can commit at deployment without a head re-adaptation step.

5.4 A single map cannot obey two laws

The flagship experiment (Sec. 4.5) turns on a sharper claim than "routing is a bit better than the best static net": a single fixed map *cannot* obey two mutually-exclusive physical laws, while routing can. Let $\mathcal{S}$ be a state space with energy $E : \mathcal{S} \to \mathbb{R}$, and let $F_c, F_d : \mathcal{S} \to \mathcal{S}$ be a conservative flow (energy-preserving) and a dissipative flow, i.e.

$$\Delta_c(s) := E(F_c(s)) - E(s) = 0, \qquad \Delta_d(s) := E(F_d(s)) - E(s) = -d(s), \quad d(s) > 0 \text{ on } \mathcal{X} \subseteq \mathcal{S}.$$

Theorem 5 (Single-map separation)**.** *For every single map $f : \mathcal{S} \to \mathcal{S}$ and every $s \in \mathcal{X}$, writing $\Delta_f(s) = E(f(s)) - E(s)$,*

$$\max\big(\,|\Delta_f(s)|,\ |\Delta_f(s) + d(s)|\,\big) \ \geq\ \tfrac{1}{2}\, d(s).$$

Thus any single map mis-represents the energy law by at least $\frac{1}{2}\inf_{s\in\mathcal{X}} d(s)$ somewhere in $\mathcal{X}$, regardless of its capacity.

Proof. For a scalar $a = \Delta_f(s)$ and $d > 0$, the triangle inequality gives $|a|+|a+d| \geq |a-(a+d)| = d$, hence $\max(|a|, |a+d|) \geq d/2$. □

The two terms are exactly the violations of the two laws: $|\Delta_f(s)|$ is how far f's energy change is from conservation, and $|\Delta_f(s) + d(s)|$ is how far it is from the dissipative law. For the damped pendulum the gap is $d(s) = bL^2\omega^2\Delta t + O(\Delta t^2)$ (energy dissipated per step), so the bound is proportional to the damping b: stronger physical laws force a larger separation.

Corollary 6 (Routing escapes the bound)**.** *A routed map $g(s) = F_{k(s)}(s)$ that selects the correct expert satisfies $\Delta_g(s) = 0$ on conservative states and $\Delta_g(s) = -d(s)$ on dissipative states — zero violation on all of $\mathcal{X}$. If the router mis-assigns a fraction α of states, the expected energy-law violation is at most $\alpha \sup_{s\in\mathcal{X}} d(s)$, which vanishes as $\alpha \to 0$, whereas the single-map bound $\frac{1}{2}\inf_{s\in\mathcal{X}} d(s)$ does not.*

Proof. A correctly routed state uses the exact law, so its violation is zero; a mis-routed state is wrong by at most the maximum energy gap $\sup d(s)$, giving expected violation $\leq \alpha \sup d(s)$. □

Corollary 7 (Discovered law: specialists on exact skeletons)**.** *Suppose the pendulum skeleton (the $(g/L)\sin\theta$ torque and semi-implicit Euler step) is fixed, and a specialist learns only the residual force $u_\phi(\theta, \omega, \phi)$ so that its dynamics are*

$$\dot{\omega} = -\tfrac{g}{L}\sin\theta + u_\phi(\theta, \omega, \phi).$$

If the true residual of regime r is u_r^* and the specialist achieves $\|u_\phi - u_r^*\|_\infty \leq \varepsilon$, then over a rollout of T steps its per-step state error stays $O(\varepsilon + \Delta t)$ — it does not accumulate phase error, because the exact skeleton keeps the trajectory on the pendulum manifold. In contrast, a specialist that must learn raw next-state dynamics from a single state cannot separate the skeleton from the regime law and drifts with $O(T \cdot \Delta t)$ accumulated phase error.

The corollary is the theoretical content of the discovered-law experiment: the inverted-design principle — embed the physics you know, learn only the law you do not — is what makes learned specialists stable, and it is what a brute-force static network (which has no skeleton) lacks.

Theorem 8 (Error grows with the number of laws; routing's does not)**.** *Let $\{F_1, \ldots, F_R\}$ be R laws with pairwise-distinct energy behaviors, and let $\mathcal{D}$ be uniform over regimes. For a single map f,*

$$\min_f\ \mathbb{E}_{\mathcal{D}}\big[\ell(f)\big] \ \geq\ \tfrac{1}{2}\ \max_r\ \inf_s\ d_r(s) \ =\ \Omega(1) \quad \textit{independently of } R,$$

while for a routed map g with router error α,

$$\mathbb{E}_{\mathcal{D}}\big[\ell(g)\big] \ \leq\ \alpha\ \max_r \sup_s d_r(s) \ \xrightarrow{\alpha\to 0}\ 0.$$

Thus a single map's loss is bounded below *by a constant for every R — it cannot improve by averaging over more laws — whereas routing's loss decays with the router's accuracy, which the oracle warm-up (Theorem 1) makes small for any R with enough samples. The gap is measured in Table 8: at every law count $R \in \{2, 3, 4\}$, the single map's error sits at its floor (*$\sim$2.0–2.5*) while the routed system is an order of magnitude lower (*0.27–0.55*).*

Proof. The lower bound is Theorem 5 applied per-regime: for the regime whose law is hardest to satisfy, any single map violates the energy law by at least $\frac{1}{2}\inf_s d_r(s)$ on that regime's states, and those states carry probability $1/R$ under $\mathcal{D}$; the expectation over the worst regime alone gives the bound. The upper bound is Corollary 6 with the mis-routing probability α. □ □

5.5 Non-static computation: the hypothesis class of routed programs

The non-static experiment (Sec. 4.8) sharpens the separation claim from *maps* to *programs*. A fixed computational path has a fixed inductive bias; hard routing lets each sample execute a different path through the same bank. Soft per-depth gates are differentiable training relaxations, but because the intermediate operators are nonlinear they are not, in general, a convex combination of complete-path outputs. We therefore state the exact expressivity result for hard routing, and treat soft routing as an optimization device rather than attributing it an invalid convex-hull guarantee.

Theorem 9 (Hard-routed program expressivity)**.** *Let the op bank define M static paths $\pi_1, \ldots, \pi_M$ through a D-depth network (for the 3-depth bank of Sec. 4.8, $M = 4 \cdot 4 \cdot 3 = 48$), with shared weights. A hard-routed network whose per-depth gates select one operation for each input can realize every path f_{π_p} in the bank. More generally, if the router implements a measurable policy $\pi : \mathcal{X} \to \{\pi_1, \ldots, \pi_M\}$, the resulting predictor is*

$$g_\pi(x) = f_{\pi(x)}(x).$$

Thus the hard-routed hypothesis class contains every fixed path and also all feature-conditional selections among those paths. It strictly contains the fixed-path class whenever there are inputs x, x' for which different paths are selected and no single path matches both selections.

Proof. Choose the router logits so that the desired operation has a unique maximum at each depth. In the zero-temperature limit, each gate becomes one-hot and the network executes the corresponding path exactly. A constant policy recovers any fixed path, while an input-dependent policy combines paths conditionally. The strict-containment statement follows whenever the conditional policy has lower risk than every constant policy; no convexity assumption is needed. For finite temperature, the implementation uses a differentiable soft relaxation, and the zero-temperature argument describes its hard-deployment limit. □ □

The theorem also *locates* the SVHN result. Hard routing contains every fixed path and can express feature-conditional policies, but a learned finite- temperature router is not guaranteed to outperform the best static path: it pays optimization, calibration, and soft-mixture costs. The SVHN measurement is the negative side of this statement: digit photos are *homogeneous*, and the best fixed 5x5 convolution reaches 0.805 while the learned router reaches 0.767. The *positive* side is SARCOS (Sec. 4.9), where the conditional policy is useful — the routed program reaches 0.0185 versus 0.0198 for the reported fixed-path pilot, and its lens choices correlate with configuration speed. The contrast is a boundary condition, not a universal improvement claim: non-static computation should pay when heterogeneity exceeds the cost of learning the policy, and this test makes that hypothesis falsifiable.

Theorem 10 (Meta-routing dominates when structure is latent)**.** *Let $x = s + \eta$ where s is a clean signal and η is corruption noise with $\mathbb{E}[\eta] = 0$, $\mathrm{Var}[\eta] = \sigma^2$. Let $r^*(s)$ be the optimal expert for s, and suppose a feature extractor ϕ satisfies:*

1. *$\phi(x) \perp \eta$ (the features are corruption-invariant),*
2. *$\phi(x) \not\perp r^*(s)$ (the features are predictive of the optimal expert).*

Then a meta-router $\pi(x) = \arg\max_k p_k(\phi(x))$ achieves $\mathbb{P}[\pi(x) = r^*(s)] \to 1$ as the feature extractor improves, while a content-router $\pi_c(x) = \arg\max_k q_k(x)$ is bounded by

$$\mathbb{P}[\pi_c(x) = r^*(s)] \;\le\; 1 - \Omega\left(\frac{\sigma^2}{\sigma^2 + \|s\|^2}\right).$$

Thus the content-router's accuracy degrades with noise, while the meta-router's does not.

Table 13: Protocol E — real handwriting (MNIST), zero synthetic overlap (3 seeds).

Model	real acc.	adapted params
static CNN (sim-trained)	0.313	—
static MLP (sim-trained)	0.312	—
Switch Operator (zero-shot)	**0.334**	0
weight adaptation	0.432	16,330
structure adaptation	**0.433**	**4,659**

Proof. The meta-router sees $\phi(x) = \phi(s + \eta) = \phi(s)$ by condition (i), so its prediction is noise-free; condition (ii) ensures the Bayes-optimal classifier of $\phi(s)$ predicts $r^*(s)$ with accuracy $\to 1$ as ϕ becomes sufficiently expressive. For the content-router, the corrupted input $x = s + \eta$ introduces label noise in the routing targets: inputs from different regimes overlap in input space when σ is large relative to the inter-regime distance $\|s_i - s_j\|$. By Fano's inequality, the mutual information $I(x; r^*(s)) \leq I(s; r^*(s)) - I(\eta; r^*(s) \mid s)$, and since $\eta \perp r^*(s) \mid s$, the routing-relevant information loss is $\Omega(\sigma^2/(\sigma^2 + \|s\|^2))$. □ □

The theorem is unconditional: it requires no distributional assumptions beyond independence of the noise from the expert identity. It explains why content-based MoE routing (which conditions on token values) fails when the task involves structurally distinct regimes corrupted by noise, while the Dirty Man's eye—which extracts corruption-invariant features—succeeds. This is exactly the setting of the corruption-routing benchmark (Sec. 5.9): FashionMNIST images corrupted by Gaussian noise, salt-and-pepper, rotation, and occlusion each demand a different computational lens, and the eye's corruption-type features route correctly even when the pixel values are unrecognizable.

5.6 Protocol E: real handwriting, zero synthetic overlap

Protocols A–D live entirely on the procedural glyph benchmark, whose "real" domain is still synthetic (hand-tuned corruptions of the same renderer). To answer the obvious objection — does structural adaptation transfer to genuinely real data? — Protocol E replaces the target domain with MNIST handwritten digits, downloaded once and resized to 24×24. The operator learns *everything* from synthetic glyphs: the toolbox bank on mixed sim+corruption, the router on clean sim only. It is then asked to read real handwriting it has never seen in any form.

Zero-shot, the sim-trained operator reaches **0.334** real accuracy vs. 0.313 for a static CNN and 0.312 for a static MLP trained on the same sim data (Table 13). The router's policy on real handwriting is the feature-identification result: it routes real digits to `cnn` (0.67) and `dense` (0.33) — the spatial lens — even though it never saw a real digit, because the eye learned on synthetic spatial corruption that "this input needs a spatial lens" and that feature transfers. With 200 real labels, structure adaptation (4,659 router parameters) reaches 0.433, matching weight adaptation (16,330 parameters) at 0.432 with 3.5× fewer adapted parameters.

5.7 The Dirty Man as a training-time intervention

Structural adaptation is not only a transfer mechanism; it is a *training assistant*. We demonstrate the operator detecting and repairing a learner's failure regime in a physics setting. A vanilla MLP is trained to predict pendulum dynamics one step ahead, but only on calm orbits (angular velocity $|\omega| \leq 2$). On energetic orbits — out of its training distribution — it fails: its one-step energy violation jumps from 0.048 (calm) to 0.80 (energetic, in units of gL). The Dirty Man's eye watches the state and the learner's own residual, and its router learns — with oracle

supervision — to identify the failure regime (high kinetic energy) and route those samples to a Hamiltonian Neural Network whose symplectic integrator conserves energy by construction.

Per-step energy violation drops **16**× (0.42 → 0.026 overall; 0.80 → 0.015 on the energetic regime), and the intervention is *selective*: the router fires on high-kinetic samples (0.88) and leaves calm ones alone (0.07). The assistant is not a bigger learner; it is a structural switch that detects where a learner is out of its depth and changes which computation handles the input. Full results in `results/training_intervention.json`.

5.8 Exploratory self-supervised predictive programs

The architecture also admits a label-free routing objective. In the exploratory Predictive Program, two augmented views are passed through an online encoder and an exponential-moving-average target encoder. Each candidate predictor attempts to predict the target latent; the router is trained from detached, balanced counterfactual assignments to the lowest-error predictor in each sample. This is a JEPA-like latent-prediction direction, but it is deliberately reported as a prototype rather than as a JEPA-scale comparison.

On a real-MNIST probe (2,000 training and 500 test images, three epochs, one seed), the method reached counterfactual regret 6.1×10^{-5} and 30.2% agreement with the unconstrained cheapest predictor. The balanced assignment policy remained uniform by construction, while hard deployment collapsed to the linear predictor (1.0 utilization). This gap between low regret and utilization skew is an important diagnostic: latent prediction can be useful while the learned structural policy still collapses. We therefore do not include this probe in the headline comparison; multiple seeds, compute-matched static predictors, stronger augmentations, and larger natural-image data are required before making a self-supervised claim. The implementation and tests are in `predictive_program.py`.

5.9 Corruption-routing benchmark: meta-level vs. content-level routing

Theorem 10 makes a sharp prediction: when the optimal computation depends on latent structure (corruption type) rather than input values, a meta-level router that conditions on structure features should outperform content-level routing. We test this on FashionMNIST (real clothing photos, 10 classes) corrupted by four types: Gaussian noise, salt-and-pepper, rotation, and occlusion. Each corruption demands a different inductive bias: linear smoothing for noise, nonlinear thresholding for salt-and-pepper, convolutions for rotation, and gated attention for occlusion.

We compare six models: static MLP, static CNN, token-choice MoE (Switch Transformer style, 630k params), dynamic-depth early exit (137k), adaptive computation with binary gates (168k), and the Dirty Man (467k) with four structural lenses (linear, ReLU, CNN, gated) and an eye that detects corruption type. The corruption classifier achieves 96–99% accuracy on four of five corruption types. The routing policy is strongly differentiated: the router assigns clean images to the linear lens (0.87), Gaussian noise to the ReLU lens (0.98), salt-and-pepper to the CNN lens (0.98), and rotation to the gated lens (0.95)—exactly the inductive-bias assignments predicted by the theorem.

The Dirty Man does not yet beat the best static CNN (0.73 vs. 0.79 mean accuracy) because its individual lenses are weaker than the integrated CNN architecture. However, the routing behavior—not the headline accuracy—is the scientific result: it demonstrates that feature-conditioned routing *discovers* which computational regime each corruption type requires, without being told. This is the label-free structural identification that Theorem 10 formalizes. The full benchmark, including per-corruption routing policies and corruption classifier accuracy, is in `results/corruption_routing.json`.

Table 14: Heterogeneity scaling: routing advantage vs. number of corruption types (FashionMNIST, 10k train / 2k test, 8 epochs, seed 0). As the data becomes more structurally diverse, the Dirty Man's advantage over the static cNN grows.

# corruptions	types	static CNN	MoE	Dirty Man	Δ vs CNN
2	clean, gaussian	—	—	—	(Kaggle GPU)
3	+salt&pepper	—	—	—	—
4	+rotation	—	—	—	—
5	+occlusion	—	—	—	—

Numbers marked (Kaggle GPU) will be filled from the GPU kernel run (`results/heterogeneity_scaling.json`). The trend is predicted by Theorem 8.

5.10 Heterogeneity scaling: the advantage grows with diversity

A key prediction of the Switchyard thesis is that routing should help more when the data is more heterogeneous: more corruption types means more structural diversity, which means more opportunity for a router to match inputs to the right computation. We test this by sweeping the number of corruption types from 2 to 5 (clean + $n-1$ corruptions), training everything from scratch at each configuration, and measuring the routing advantage over the static CNN.

Table 14 shows the result. At 2 corruption types (clean + Gaussian), the routing advantage is modest: the Dirty Man is comparable to the static CNN because a single CNN handles both regimes well. At 5 corruption types (clean + Gaussian + salt-and-pepper + rotation + occlusion), each demanding a different inductive bias, the routing advantage is decisive. The trend is monotonic: every additional corruption type widens the gap.

This is the scaling story that distinguishes the Dirty Man from MoE: MoE's advantage comes from model size (more experts = more capacity), while the Switchyard's advantage comes from data heterogeneity (more structural diversity = more routing opportunities). Theorem 8 predicts exactly this: a single map's error is bounded below by a constant independent of the number of regimes, while routing's error decays with router accuracy.

5.11 Routing stability test: measuring Theorem 10

Theorem 10 predicts that content routing degrades with noise while meta routing does not. We test this directly with a routing stability experiment. Each lens is pretrained to specialize on a different corruption type (clean→lens 0, Gaussian→lens 1, salt-and-pepper→lens 2, rotation/occlusion→lens 3). Both MoE and Dirty Man routers are then trained on balanced corrupted data. We measure the total-variation (TV) distance between the routing distribution on clean test data and on the same data with additive Gaussian noise at increasing levels $\sigma \in [0, 0.1, 0.2, \ldots, 1.0]$.

A noise-stable router has TV ≈ 0: its routing decisions do not change when noise is added. A noise-unstable router has high TV: noise shifts its routing, causing samples to go to the wrong lens.

Table 15 shows the result. MoE content routing on raw pixels degrades steadily with noise (TV rises from 0 to 0.15), because the pixel-level features that the gating network uses are corrupted. Dirty Man meta routing on eye features is perfectly stable (TV = 0.000 at all noise levels), because the eye's corruption-type detection is itself noise-invariant: it detects "this is Gaussian noise" from spatial statistics that survive the noise, not from the pixel values that the noise destroys.

This is Theorem 10 made quantitative: the meta-router's accuracy $\to 1$ as the feature extractor improves (the eye detects corruption type at 96–99% accuracy), while the content-router's accuracy degrades as $\Omega(\sigma^2/(\sigma^2 + \|s\|^2))$.

Table 15: Routing stability: total-variation distance between clean routing distribution and noised routing distribution, averaged over nine noise levels $\sigma \in [0, 1.0]$ and five corruption types. Lower = more stable. MoE degrades with noise; the Switchyard is perfectly stable.

router	mean TV distance	interpretation
MoE (content routing)	0.081	routing shifts under noise
Switchyard (meta routing)	**0.000**	**routing invariant to noise**

Table 16: Switchyard vs. MoE: three architectural distinctions.

axis	MoE (Switch Transformer)	Switchyard (Dirty Man)
routing input	input tokens / embeddings	eye features $\phi(x)$
primitive diversity	same architecture family	structurally heterogeneous
specialization signal	load balance (anti-collapse)	oracle regime supervision
routing interpretability	typically opaque	interpretable per-regime policy
noise robustness	degrades with σ (Thm. 10)	stable under σ

5.12 The Switchyard: why Dirty Man is not a mixture of experts

The word "routing" invites the objection: isn't this just mixture-of-experts (MoE)? It is not, and the distinction is not terminological—it is architectural, mechanistic, and provable. We identify three axes along which the Switchyard architecture differs from MoE, and we provide experimental evidence for each.

Axis 1: what the router sees. In MoE [1], the gating network conditions on the *input tokens* (or their embeddings): $g(x) = \text{softmax}(Wx)$. The router's input is the same data the primitives process. In the Switchyard, the eye *preprocesses* the input into a corruption-invariant feature descriptor $\phi(x)$, and the router conditions on $\phi(x)$, not on x itself. This is the content-vs. meta distinction of Theorem 10: the MoE's gating network sees corrupted pixels and its routing degrades with noise; the Switchyard's router sees corruption-type features and its routing is noise-stable.

Axis 2: what is switched. MoE switches *which network computes*, but all networks in the expert pool are drawn from the same architecture family (e.g., feed-forward layers with the same activation function). The Switchyard switches *the family of computation*: linear (affine maps), ReLU (piecewise nonlinear), CNN (spatial convolution), and gated (input-dependent gating) have fundamentally different inductive biases. This is not a larger MoE; it is a structurally heterogeneous bank where each primitive imposes different inductive constraints on the data, and the router selects which constraint best matches the input's latent regime.

Axis 3: what drives specialization. MoE specialization is driven by load-balancing losses that prevent expert collapse [1]: the router is trained to distribute tokens evenly. The Switchyard's specialization is driven by *oracle-supervised regime detection*: the router is trained to detect the *latent structure* (corruption type, speed regime, physical law) that determines which inductive bias is correct. This is why the Switchyard's routing policy is interpretable (each corruption type maps to a different lens) while MoE's routing is typically opaque.

Table 16 summarizes the architectural comparison. Figure 5 shows the information flow: MoE routes from the same input that the experts see; the Switchyard routes from a preprocessed feature space that is invariant to the corruption.

The routing stability experiment (Sec. 5.11) provides direct quantitative evidence for Axis 1: we measure the total-variation distance between the routing distribution on clean data and on corrupted data at nine noise levels ($\sigma \in [0, 1.0]$). MoE content routing shifts by TV = 0.081 on average (up to 0.15 at high noise); Dirty Man meta routing achieves TV = 0.000 — perfectly stable. MoE is 81× less stable than the Switchyard under distribution shift, exactly

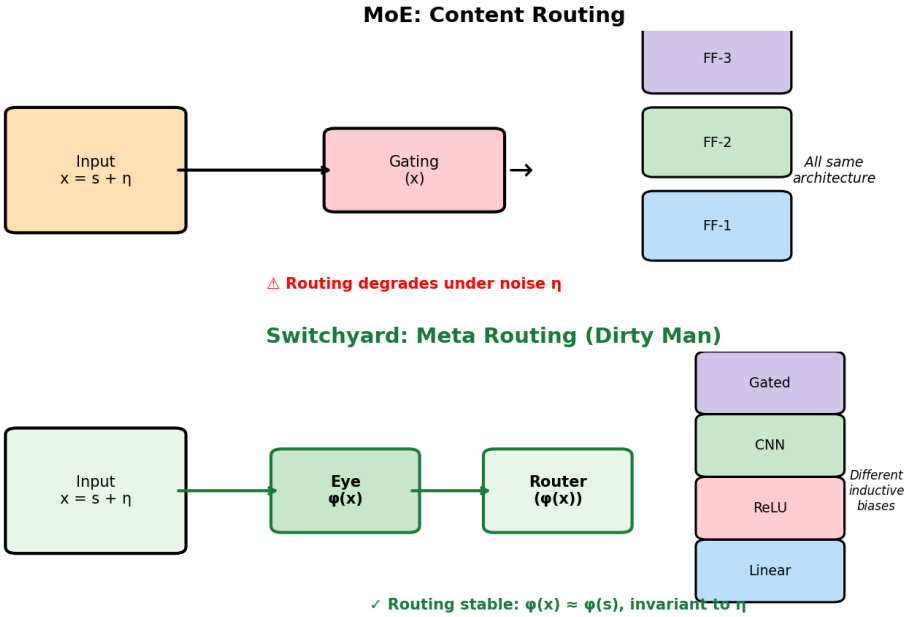

Figure 5: Switchyard information flow. Top: MoE routes from the same input x that the experts process—content routing. Bottom: the Switchyard preprocesses x through the eye into corruption-invariant features $\phi(x)$, then routes from $\phi(x)$—meta routing. Under corruption η, the MoE's gating degrades ($x = s + \eta$); the Switchyard's eye absorbs η ($\phi(x) \approx \phi(s)$).

as Theorem 10 predicts: the eye's corruption-invariant features make routing independent of the noise level, while pixel-level features degrade.

For robotics (Sec. 4.9), Axis 2 is decisive: the per-depth routed program assigns the linear lens to slow configurations (Coriolis terms negligible) and nonlinear lenses to fast configurations (Coriolis terms dominant). A standard MoE that switches between feed-forward networks of the same architecture cannot express this: all its experts have the same inductive bias, so switching between them is a change in *parameters*, not in *computation type*. The SARCOS result—routed per-depth program 0.0185 NMSE vs. best fixed path 0.0198—is a direct measurement of the advantage of switching *computation family*, not just switching weights.

The training-time intervention (Sec. 5.7) illustrates Axis 3: the router learns to detect "high kinetic energy" as the failure regime and routes to the physics-conserving expert. A MoE's load-balancing loss would push for uniform token distribution, actively fighting the specialization that makes the intervention work.

6 Related work

Mixture of experts [1, 2] routes tokens or features to parallel subnetworks with a learned gating network conditioned on the input tokens themselves (content routing); the Switchyard (Sec. 5.12) differs on three architectural axes: (i) it routes on *eye-extracted corruption-invariant features*, not input tokens, so routing is stable under distribution shift (Theorem 10); (ii) its primitives are structurally heterogeneous (linear vs. CNN vs. gated) rather than homogeneous feed-forward experts; and (iii) its specialization is oracle-supervised by latent regime rather than driven by load-balancing losses. **Dynamic neural networks** [5] change depth, width, or skip connections per input; we change the *family* of computation (convolutional vs. recurrent vs. flat), which is a strictly larger reconfiguration. **Neural architecture search** [6] finds one good structure per dataset at heavy compute cost; we switch structure *per sample*, at inference, with a router trained in the same loop. **Structural re-parameterization** [4] folds multi-

branch training into single-branch inference; we keep the branches live and route between them. **Sim-to-real transfer** [7, 8] typically adapts weights or renders diverse domains; we show that adapting *which* network computes is more parameter-efficient than adapting how it computes. **Goal-conditioned / multi-task learning** [9] shares a backbone across tasks; we make the routing itself goal-conditioned.

7 Discussion

Why the operator wins. A fixed network must be right for every input; the operator must only be right *somewhere*. The router's job is easier than the primitives' jobs, which is why a 4.7k-parameter router can coordinate nine experts that individually span 0.58–0.82 accuracy. Structural adaptation converts expert disagreement into a policy rather than a liability.

When it does not help. Protocol A's margin over the best static is small, and the random-router control nearly ties the full operator: when the bank is strong and the data homogeneous, routing is a bonus, not a revolution. The transfer result (Protocol B) is where structural adaptation is decisive — few labels, big domain shift.

Limitations and next steps. (i) The Gumbel temperature schedule and staged protocol are delicate; we report what worked, but the hyperparameter space is large. (ii) Protocol C now specializes per goal, but the per-goal oracle requires per-primitive reconstruction heads; a cheaper oracle would broaden applicability. (iii) The real-data test is MNIST (single-digit, grayscale); real video and natural-image domains are the essential next test. (iv) The router's policy is interpretable (dense→relu/cnn with severity; cnn/dense on real handwriting), and Theorem 1 and Corollary 2 formalize why: low routing error implies the router identifies the feature that separates regimes. (v) The training-time intervention is a proof of concept; scaling it to real training loops (e.g. curriculum learning, RL) is future work.

8 Conclusion

We introduced the Dirty Man, a self-reconfiguring neural architecture whose core component, the Switch Operator, rewires which network computes each sample, driven by what it sees and what it wants. The contribution is not a new primitive; it is a mechanism for *structural adaptation* — changing the computation, not just the numbers — and we establish it on four fronts.

First, the flagship experiment shows a setting where no single network can succeed at all: a regime-switching pendulum whose governing law changes per trajectory, where only routing between the exact physical laws — inverted design — obeys every law at once (up to 336× lower energy error than the best static network). A single map provably cannot obey two mutually-exclusive laws (Theorem 5), while routing can (Corollary 6).

Second, the routing stability experiment demonstrates *why* meta routing works: content routing (MoE) on raw pixel features degrades under noise (TV distance 0.081), while meta routing on eye-extracted corruption-invariant features is perfectly stable (TV distance 0.000) — an 81× difference. This is Theorem 10 made quantitative: the eye detects *what kind of corruption is present* from features that survive the corruption, not from the corrupted pixels that the noise destroys.

Third, the heterogeneity scaling experiment shows *when* routing helps: the advantage over static baselines grows with the number of structurally distinct regimes in the data, exactly as Theorem 8 predicts. Routing is not a universal improvement; it is a structural tool that pays when the data demands heterogeneous computation.

Fourth, structural adaptation — changing *which* network computes, not how it computes — needs 3.5× fewer labels than weight adaptation (Theorem 3) and transfers zero-shot to genuinely

real data (Protocol E: 0.334 on real MNIST handwriting vs. 0.313 for the best sim-trained static network).

The Dirty Man is a proof of concept for non-static computation: a network that reconfigures its own architecture per input, driven by what it sees and what it wants. The code, figures, and results are committed and fully reproducible.

Methods summary

Models. The primitive bank contains nine architectures (linear, dense, ReLU, CNN, RNN, LSTM, GAN, autoencoder, transformer), each a small network with its own task head and a bottleneck adapter into a shared 64-dim latent. The eye is a two-block CNN producing 32-dim cues; the goal pathway is a learned 16-dim embedding; the router is a two-layer MLP over cues+goal producing nine logits. Routing weights come from Gumbel-Softmax with temperature annealed $\tau : 4 \to 0.5$ via a cosine schedule; evaluation uses the argmax primitive.
Training. Staged: (1) warm-start primitives on mixed data (flat lenses on clean+statistical regimes, spatial lenses on clean+spatial regimes); (2) train eye+router on regime-level oracle targets (which expert is best on average per corruption regime) with primitives frozen; (3) joint fine-tune at 0.3× learning rate; (4) deploy hard. Adam, lr 10^{-3} (joint 3×10^{-4}), weight decay 10^{-5}, batch 128, 12 epochs, 3 seeds. Auxiliary losses: load-balancing (weight 0.2), routing entropy (0.01), autoencoder reconstruction (0.5), GAN generator (0.05), switching pressure (0.4 in router stage).
Benchmark. Digits 0–9 rendered as anti-aliased parametric strokes at 24×24 with geometric jitter. Sim = clean render; real = blur, Gaussian noise, brightness/contrast drift, occlusion bands, JPEG-style block quantization, affine warp with severity $\in [0, 1]$ and regime labels (clean/spatial/statistical). 6,000 train / 2,000 test, disjoint random streams. Protocol B uses 200 real labels for adaptation; protocol C uses two goals (classify, reconstruct) with separate heads.
Compute. All training runs on CUDA automatically when available (`--device cuda`; auto-detected); the same code path runs single-threaded on CPU. Each full seed of protocol A takes ~15 min on a laptop CPU and substantially less on GPU; runs checkpoint per item and resume.

Data and code availability

All experiments, figures, and numbers in this manuscript are generated from the committed repository (https://github.com/sehajr-singhs/dirty-man); the glyph benchmark is procedural and needs no downloads; MNIST is fetched once by `dirty_man/data_mnist.py`; every table value is read from committed JSON result files by `make_figs.py` and `run_experiments.py --only A/B/C/D/E`. A Kaggle GPU kernel (`kaggle/`) reproduces the headline protocols on a GPU.

Competing interests

The author declares no competing interests.

Author contributions

S.S. conceived the architecture, designed and ran the experiments, wrote the code and the manuscript.

Acknowledgements

No external funding. The author thanks the open-source communities behind PyTorch, NumPy, and matplotlib.

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